Title – Enzyme Required Practical

What enzyme is present in saliva?
What does this enzyme break down?

Monday, 11 December 2023

Enzyme Required Practical

Learning objectives:

By the end of the lesson you will be able to:

- describe how safety is managed, apparatus is used and accurate measurements are made
- explain how representative samples are taken
- make and record accurate observations
- draw and interpret a graph from secondary data using knowledge and observations.

Key Words

Amylase

Iodine Solution

Starch

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CONNECTION

Complete the BBC Bitesize revision Due 5th October

Aim - To investigate the effect of pH on the rate of digestion of starch by amylase

Complete a prediction, which pH will digest the starch the fastest?

Watch the demo

Write out a method maximum of 6 bullet points

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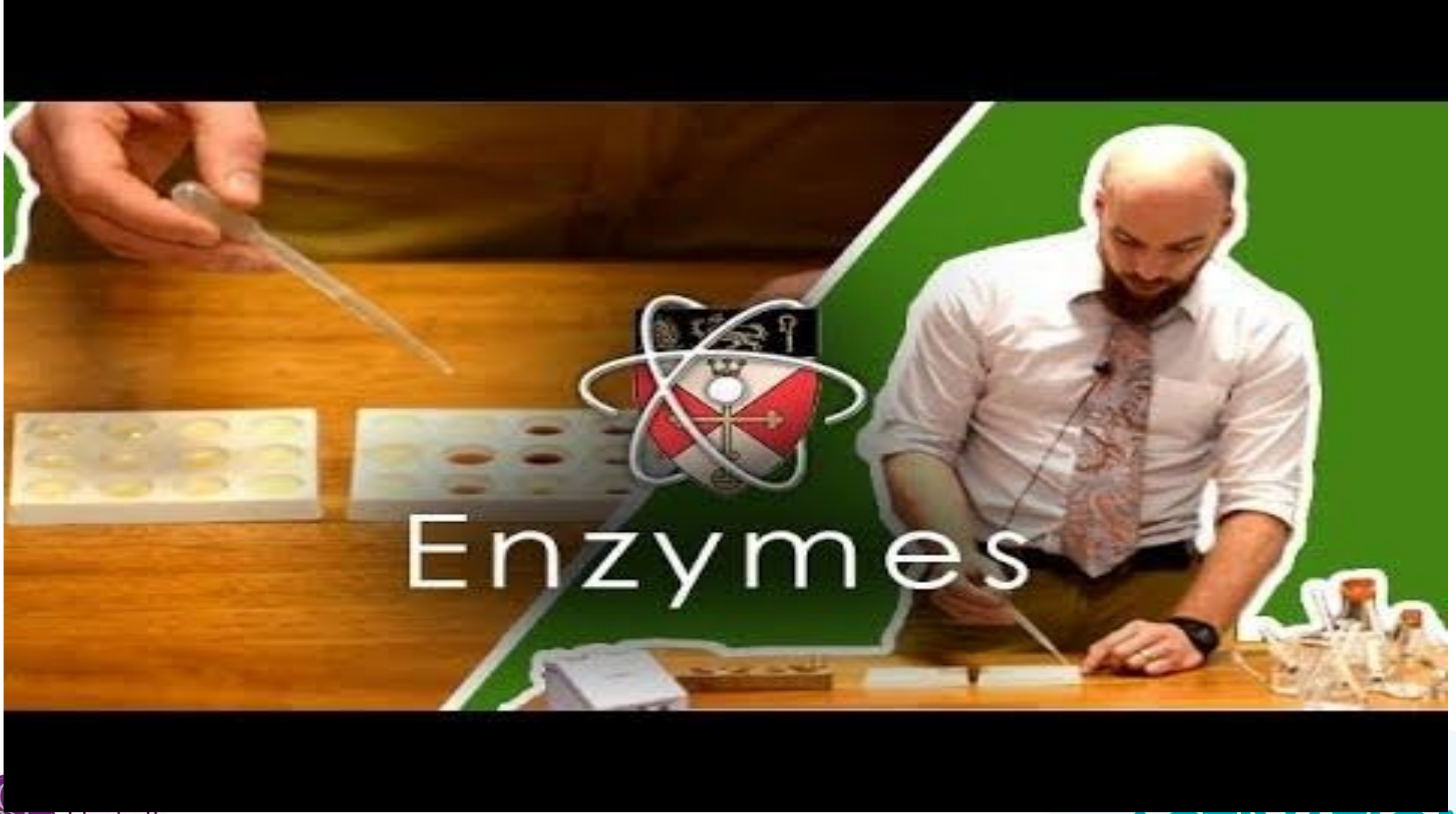
ACTIVATION



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ACTIVATION

Write out a method maximum of 6 bullet points

- Add one drop of iodine to each of the wells in the spotting tile.
- Use a pipette add 1cm^3 of amylase to a test tube
- With a second pipette add 2cm^3 of the chosen pH into the same test tube as the amylase
- With a third pipette add 2cm^3 of starch solution to the same test tube and mix well with a pipette.
- After 20 seconds add a small amount of the mixture to one of the wells in the spotting tile. Put the rest of the solution in the pipette back into the test tube.
- Repeat the previous step every 20 seconds for a maximum of 40minutes
- If there is time repeat with a different pH

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CONSOLIDATION

pH	Time (s)
2	240
5	170
7	80
8	150
10	220

1 Explain why amylase, starch, and iodine are used in this investigation.

2 Explain why is it important that the concentration and volume of all the test starch solutions and the enzyme added are known and are the same.

3 Explain why all the test solutions are kept in a water bath at the same temperature and why that temperature must be controlled below 37 °C.

4 Explain the purpose of the spotting tiles with iodine in the wells.

5 The pipettes used to take samples must be rinsed out with clean water between each sample. Suggest a reason for this.

6 If all the starch is broken down before the first sampling, or if no starch was broken down after an hour, it would be hard to get useful results. Suggest reasons for both of these situations and ways of overcoming the difficulties.

1. Iodine is an indicator for starch, amylase breaks down starch.
2. These variables need to be kept consistent across the experiments so the results can be compared. We want to compare how pH affects the enzyme, nothing else.
3. To mimic the natural temperature of the body, this will give the optimum temperature for the enzyme to work.
4. A white background will make it clearer for us to see the colour change, iodine will change colour if starch is present.
5. To prevent cross contamination. So a different pH buffer is not added which will make the experiment invalid.
6. We would not be able to quantify the results so the pH may need to be changed so that we are able to see a change.

Make sure to collect any results you produce.
Complete your result table using other peoples results for different pH's

Create a graph using your results. (**state** which graph you will use and explain **why**)

Conclude your results, use the data you have collected to back up you prediction and your conclusion.

pH	Time (s)
4.5	
5	
6	
7	
8	

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CONSOLIDATION

- Plenary

  - Discuss your findings of your practical with your table – where they what you expected?

  - What were some of the limitations?

  - What could have been improved?

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